APPLICATION BRIEF

NGS Application

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Automating NGS Sample Prep with the Zephyr NGS Workstation

Introduction

The Zephyr® NGS Workstation is a benchtop liquid handler designed to automate preparation of libraries for next generation sequencing and to address the growing number of laboratories with requirements for the physical separation of pre-and post-PCR processes. The simplified user-interface, pre-programmed NGS library protocols, and integrated hardware maximize laboratory productivity while reducing variability resulting from manual pipetting steps.

Laboratoire CERBA separated their NGS library preparation workflow into pre- and post-PCR processes with two Zephyr NGS Workstations. Their focus is to simultaneously detect 139 clinically relevant cystic fibrosis disease-causing mutations and variants of the cystic fibrosis transmembrane conductance regulator (CFTR) gene from genomic DNA samples.

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Materials and Methods

Automation of sample preparation followed the standard workflow specified by the NGS reagent kit vendor. The workflow was partitioned into five segments, starting with genomic DNA and ending with normalized NGS library prepared samples to be loaded onto an Illumina MiSeq®DX sequencer.

Protocols where divided between the pre- and post-PCR Zephyr NGS workstation instruments (Figure 1). Dead volumes were minimized through the optimization of pre-programmed protocols.

For this evaluation, 48 samples were processed manually, while the same replicate 48 samples were processed using the Zephyr NGS. Unlike with manual methods, the automated processing of 48 samples could be processed in parallel. A quality control check was performed post-PCR cleanup using the DNA High Sensitivity Assay on the LabChip® GX instrument (Figure 2). Following normalization, samples were loaded onto the Illumina MiSeq®DX. Sequencing results demonstrate 100% correlation between manually and automated sample preparation workflows; thus, underscoring increased efficiency with no loss in sample quality (Figure 3).

Results and Discussion

The Zephyr NGS produces sequence-ready samples with improved reproducibility and consistency while minimizing the labor-intensive steps associated with manual NGS sample preparation.

With a walk-away solution and flexible sample processing (8 to 96 samples), the Zephyr NGS is an ideal solution for medium throughput labs enabling the parallel processing of 96 samples in less than eight hours.

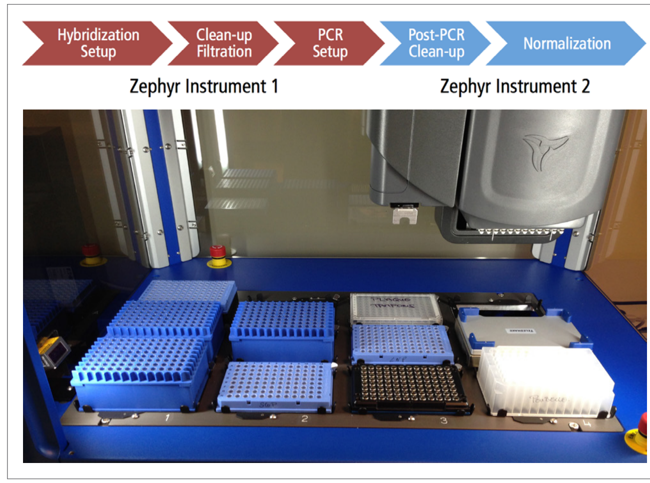

Figure 1. Zephyr NGS deck setup. Complete sample preparation for up to 96 samples can be achieved in less than eight hours.

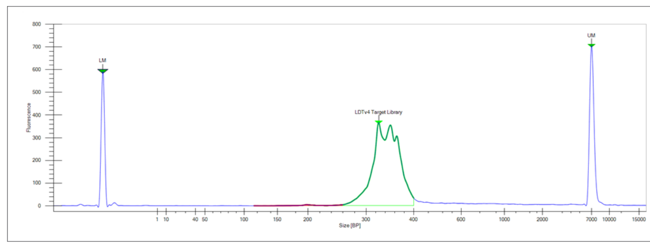

Figure 2. Electropherogram from LabChip GX Instrument showing a single sharp peak representing the sample library and no contaminating or residual DNA fragments.

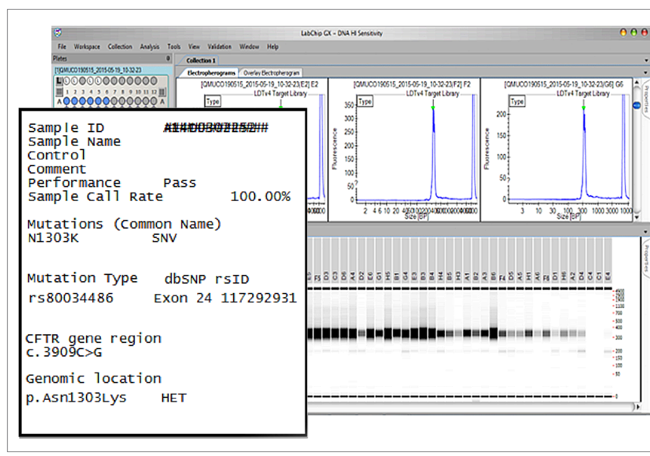

Figure 3. NGS library preparation sample processing comparison. LabChip GX data (screenshot) and the Illumina MiSeq®DX report indicate complete correlation between manually and automated sample preparation processes. This highlights the increase in throughput and efficiency provided by automated processing workflows without compromising sample quality.

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